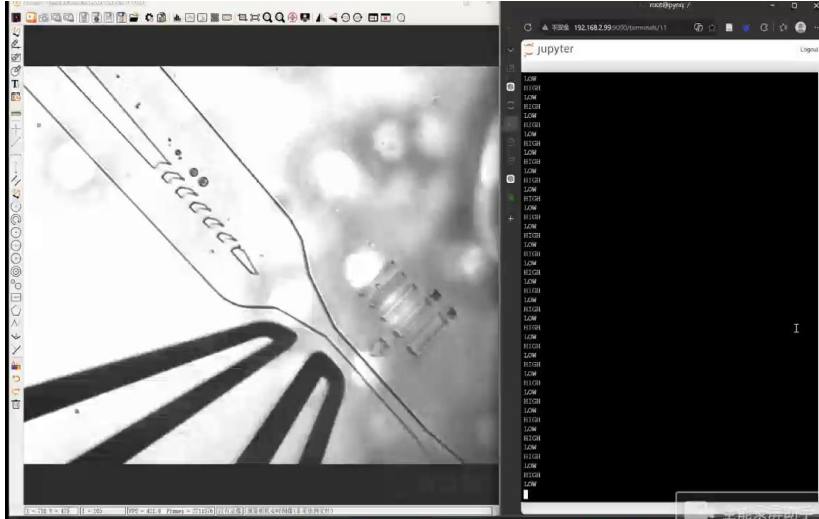


Lab Notebook (Microfluidic Platform)

September

DEP Sorting Performance



Video 14 — Sorting performance at 1 Hz

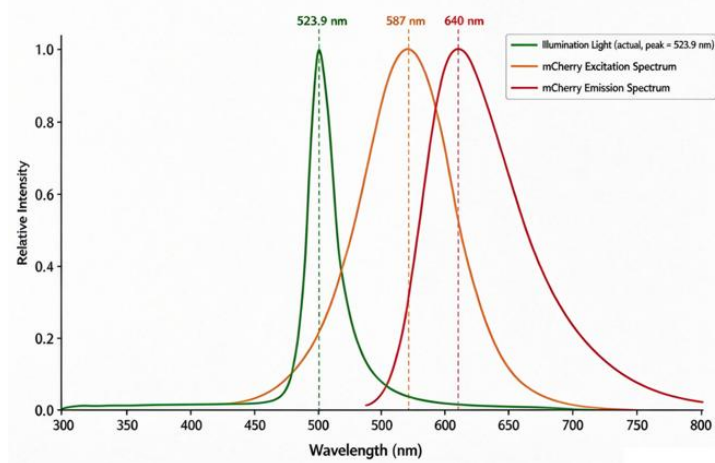
The sorting performance of the system was validated. By controlling the cyclic switching of the sorting gate, the sorting signal was toggled at 1 Hz, and droplets flowed into different channels according to the sorting instructions, demonstrating that the sorting function works stably.

Optical-Path Modification for mCherry Adaptation

This detection platform was initially built for eGFP fluorescence: eGFP is excited by a 488 nm light source, and the emitted fluorescence passes through a 515 – 555 nm emission filter before being collected by the PMT; the 660.48 nm observation illumination is blocked by this emission filter, avoiding interference with fluorescence detection.

By modifying the front-end optical path, the platform can be adapted to mCherry fluorescence detection, while the back-end signal acquisition, droplet detection, and sorting logic need no modification and can be directly reused. mCherry-channel optical modification: replace the excitation light source with one matching the mCherry excitation peak (~587 nm), and replace the emission filter with one matching the mCherry emission peak (~610 nm); the observation illumination must be re-selected so that it neither excites mCherry nor passes through the 610 nm emission filter (e.g., use short-wavelength green illumination, or keep ~660 nm illumination while confirming that the filter is a bandpass rather than a longpass).

Comparison of mCherry Excitation/Emission Spectra with Illumination Light



All videos referenced in this document are available for download at the bottom of our team's Notebook page.